

MONICA NURSING STUDY CENTER

Laboratory Tests

Study Guide

Organized review material

Prepared for study and printing

MISCELLANEOUS. Lab Test
Serum Electrolytes

Potassium = 3.5-5.0

< 3.5 Hypokalemia	> 5.0 Hyperkalemia PRIORITY
CAUSES	
- Diuretics except K sparing - furosemide↓ - Digoxin - Cushing - Steroid - Amphotericin B - Licorise - Conn disease H₂O+ Na yo= K- toilet - EKG = U wave	- Addison - Renal failure - Diarrhea, Iliostomy - Diabetes Ketoacidosis (same tto to low K to 3.5 = Dextrosa+ Insulina) - Priles, Sartan - Espirinolactona - Tumorlysis Syndrome ↑K, ↑ Uric Acid, ↑ Po₄, ↓ Ca. - Burn Injuries during first 24 hours - Burn Injury > 72hours = Low K
 Hypokalaemia T wave inversion ST depression Prominent U wave	 Hyperkalaemia Peaked T waves P wave flattening PR prolongation Wide QRS complex
Treatment K Cloruro = Infusion Pump Never IV push or IV bolus. No more than 10 meq/hr Dilute 1 meq= 10 ml of NaCl.	

EKG Hyperkalemia



- Tall T or Peak T

1. Ca + Mg + ↓ QT prolonge = Torsade de Pointe



Treatment Hyperkalemia

In Emergency situation

1. NaCl 0.9%
2. Dextrosa + Insulina
3. Sodium Polystyrene or kayexalate (oral or enema)

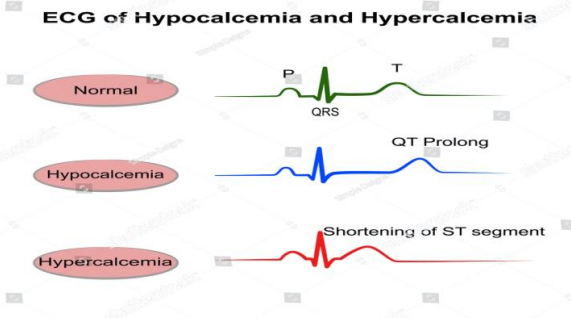
Side Effects

- ✓ Diarrhea (call Dr.)
- ✓ Increased Na

In case of Cardiac Arrhythmias

- ✓ Administer Calcium Gluconate (antidote)

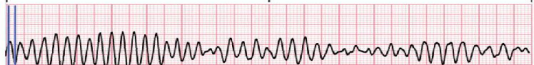
Calcio= 9-11

Less than 9 Priority Hypocalcemia <u>Cause</u> 1. Thyroidectomy 2. Hypoparathyroidism 3. Pancreatitis Chevosteck 4. Massive blood transfusion Por el citrato para no coagulation pero baja el Calcio 5. Tumorlysis 6. Renal Insuficiency 7. Metabolic Alkalosis <u>Assessment Hypocalcemia</u> 1. Paresthesias in thyredectomy = Emergency 2. Chevosteck 3. Trousseau 4. Laryngeal spam= Stridor 5. QT Prolonged= Torsade de point = Traqueostomy set + Calcium gluconate at bed side. - o Calcium Gluconate = Slowly through stable vein to prevent skin necrosis	More than 11 Hypercalcemia • If cardiac arrithmyas = Priority <u>Cause</u> 1. Osteoporosis 2. Metastasis 3. Bone Cancer (Osteosarxcoma) 4. Multiple Myeloma <u>Assessment</u> - Renal Calculi <u>Treatment</u> 1. Plenty of Fluids Acid – Arsh Diet Ex. Cranberry Juice ECG of Hypocalcemia and Hypercalcemia  shutterstock
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Sodium 135-145

Less than 135 = Priority Brain Edema = Seizure Client Na< 120 = Seizure Prevention - SIADH - Addison - Lithium - TURP Syndrome - Near drowning Fresh water (river)	More than 145 - Diabetes Insipidus - Cushing - Near drowning salt water (ocean) - Conn Disease.
<u>Treatment</u>	
1. Hypertonic solution 3% NaCl.	1. Hypotonic 0.45% 2. Client Na = 157 a) 0.45 NaCl b) 0.45 NaCl + Dextrosa (para que baje la sal)

Magnesium 1.5 – 2.5

Less than 1.5 Hypomagnesemia	More than 2.5 Hypermagnesemia
- Malnutrition - Alcoholism <u>Assessment</u> QT = Prolonged + Torsade de point	- Magnesium Sulfate Intoxication in pregnant = > 7.0 - Normal > 2.5 in pregnant 1. Respiratory depression 2. Decreased assist DTR 3. Decreased urinary output <30
<u>Treatment</u>	
- Magnesium Ventricular Tachycardia Torsade de Pointes - EKG Reference 	- Calcium Gluconate (antidote)

Phosphorus 2.5-4.5

More than 4.5 is the priority
<u>Cause</u> Renal Failure <u>Treatment</u> Aluminium hydroxide (phosphate biding) (biding = aluminium hydroxide + food or snack) Amphojel Sevelamer Renagel Where to administer with food as binding

Inflammation Markers

1. Erythrocyte Sedimentation Rate.

- Normal = 0-25
- > 30 = Inflammation

2. Protein C Reactive

➤ Hemoglobin

- Normal = 12-16 g/l
- Less than 12 = Anemia

Nota: Cornstarch = (maizena cruda) if a pregnant eats cornstarch mean

PICA = Anemia

- More than 16

- Polycythemia Vera = Chronic Hypoxemia (sangra la nariz en alturas)(excesive formation of red blood cells)
- Thrombosis = Stroke
- MI, DVT

**Treatment**

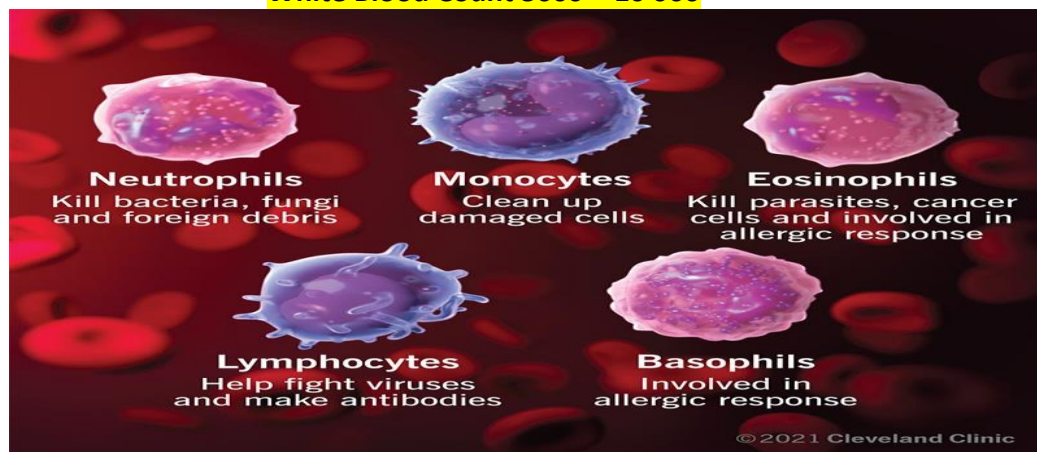
1. Acetazolamide
2. Hydroxiurea (cancer and to prevent polycythemia crisis)

Nota: Aortic Stenosis = Syncope on exertion.

Hematocrit (= 36-48 or = HB x 3)

<u>Less than 36</u>	<u>More than 48</u>
Anemia	Fluid deficit
Fluid Overload	Dehydration
<u>Nota: H2O opposite to Hematocrit</u>	
<u>Hematocrit = BUN</u>	

White Blood Count 5000 – 10 000



- Less 5000 = Immunocompromised
- **Priority less 3500 around**

Causes

- Steroids
- **Clozapine**
- Mab
- Chemotherapy
- Carbamazepine
- Propylthiuracil and methimazole (Antithyroids = Hyperthyroids)
- WBC < 3000 = Emergency
- Medication are main WBC decreased

Medication lower WBC

- Clozapine
 - Carbamazepine
 - PTU, Methimazole
 - Mab, (termination)
 - Chemotherapy
 - Steroids
- } Sore through
- } Or
- } Fever

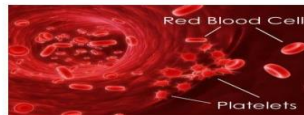
More than 10 000 WBC = Infection**➤ Except on**

- Labor = 20 000- 25 000 = normal
- 1er trimester of pregnant = 15 000 – 16 000 = normal
- Newborn
- MI
- Long Term Steroid

Eosinophils = ↑ Allergies or Asthma

Lymphocytosis = Viral infection

Leukocytosis Neutrophilia } Bacterial Infection
with shift to the left }

Platelets 150 000 – 450 000**Platelets**

- Do not contain a nucleus
- produced in bone marrow
- when blood vessels are ruptured, they initiate the **blood clotting** reactions

<http://youtube.com/watch?v=1ncvcgffGIY>

Less than 150 000 = Thrombocytopenia

1. Priority < 50 000 = Risk for spontaneous bleeding.
2. Priority < 20 000 = Platelets Transfusion (2 hours)

450 000 = Polycythemia Vera.

Clotting time 8-15 minutes

- Client should not receive heparin therapy for 3 hours before specimen collection
- More than 15 = follow up

APTT – check Heparin

- Normal value without Heparin = 25-35
- With Heparin = 45- 75 – Document
- More than 75 = Stop heparin and Call Dr. (Risk for bleeding)
- Less than 45 call Dr. to increase doses (Risk for thrombo).

Board: test to stop heparin and continuo home with warfarin.

***PT* = 9.6 – 11.8 (9-12)**

- Warfarin = Normal until 18
- Initiating Bleeding precaution if more than 30.

***INR * Normal = 1-2**

- With warfarin = 2-3
- If more than 3 – Stop, Call Dr. (Risk for bleeding)
- Less than 2 = coagulation – Call Dr to increased Doses.

Diagnostic Diabetes

1. Fasting Glucose

- Normal = 70-110 mg/dl
- Diabetes = more than 126 mg/dl
- Pre- Diabetes or Metabolic Syndrome = 110 – 126 mg/dl

2. A1C

- Normal = until 5.6
- Diabetes = more than 6.5
- Metabolic Syndrome = 5.6 – 6.5

To check 90 previous day function – A1C = 7

Less than 7 for control

Good control = <7

Bad Control = > 7

3. a) 1 hour Glucose Challenge Test for Pregnant.

- Do not need Fasting
- Administer 50 g of glucose and then check blood glucose 1 hour after.
- More than 140 mg/dl = Diabetes

b) Glucose Tolerance Test (More specific)

- Fasting Glucose
- Administer 50 g glucose = reads 1 hour
- Administer 75 g glucose = reads 1 and 2 hours
- Administer 100 g glucose = reads 1, 2 and 3 hours
- More than 140 mg/dl = Diabetes

Cardiac Studies

➤ BNP or Natriuretic Peptide

- > 100 = CHF
- > 700 end State CHF = No priority at this point

Nota: Pink Frothy Sputum = Acute Pulmonary Edema = Emergency

Cardiac Enzymes (Myocardial Infarction)

- CK- MB = Lasting elevated 72 hours.
- Troponin = **More specific diagnostic** – Lasting elevated 7-10 days.
- Myoglobin = Lasting elevated 6 hours.

Nota: difference between MI and Angina = Troponin.

Serum Gastrointestinal Studies

- Albumin 3.5 – 5.0
 - Fasting is not required
 - Less than 3.5 = Edema
 - = Nutritional Status
- Pre- Albumin 15-30
 - Less than 15 = Malnutrition = Risk for Pressure Ulcer
- Ammonia Level Normal = 10-80
 - More than 80 = Ammonia Encephalopathy
 - = LOC Alteration
 - = Seizure
 - = Asterixis = Unintentional Hand Tremors

Decreased Ammonia

- | | | |
|--|---|---|
| <ol style="list-style-type: none"> 1. Neomycin 2. Lactulose 3. Decreased Protein Diet | } | Treatment for
Encephalopathy
Ammoniacal |
|--|---|---|

Liver Enzymes

- ✓ ALT = 4-6 unit/L (Alanine aminotransferase)
- ✓ AST = 0-35 unit/L (aspartate)
- > 6 } Liver Damage. (jaundice) Hepatitis
- > 35 } Call Dr.

Pancreatic Studies

- ✓ Amylase = 25 – 151 unit/L
In acute pancreatitis increased 5-6 times normal value = 600-700
- ✓ Lipase = 10 – 140 unit/L (**More specific**)
Increased after first 24 hours lasting elevated up to 14 days
- ✓ CT Scan with or without contrast.

Bilirubin

Bilirubin Total Adult = < 1.5 mg/L Newborn = <12mg/L	Client eat Diet low in yellow food Avoid: Yams Yellow bean Pumpkin Carrots
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Lipid gram

- 8 – 12 hours **Fasting**
- No fluids except water 12 – 14 hours before
- No alcohol 24 hours before.

50 - HDL = > 50 women - > 40 males.

100- LDL	}	Normal
150 -Triglycerides		Less
200 -Cholesterol		Than

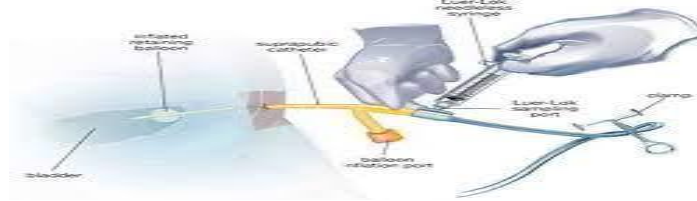
H I V Testing

1. Elisa (2 times)
2. Western Blot = Confirmatory Diagnostic
Aids
 - CD4 less than 200 (aunque haya sido solo una vez)
3. Viral load
 - Risk for Aids in HIV client.
 - < 10 000 = low risk
 - > 100 000 = Very high risk
 - For Vaginal Delivery less than 100.

Renal Exam

✓ Urinalysis

- Clean Catch Method (1st)
- Take first void in the morning
- Nitrate and leukocyte esterase positive = UTI



✓ **Urinary Culture (Confirmatory)**

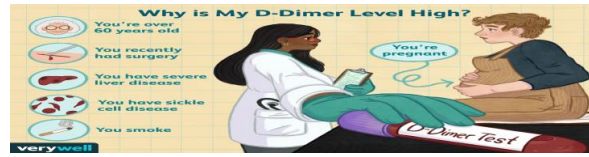
- Take Sample mid stream
- **> 100 000 u x field = UTI Positive**
- If lower repeat test.

Creatinine 0.6 – 1.3 mg/L

- Renal Insufficiency = $> 1.3 \text{ mg/dl}$
- Creatinine Clearance = more specific Test for renal insufficiency
- Blood creatinine Level + 1-3
- GFR Urinary collection 24 hours = Discard all First Void.
- GFR = less than 90.

BUN = 10-20 (fluid Status)

- ✓ > 20 = Fluid Volume Deficit = Dehydration
- ✓ < 10 = Fluid volume Overload

D. Dimer (Positive = Thrombosis)

- DVT
- STROKE
- Pulmonary Embolism (VQ Scan = Confirmatory Test)
- DIC

RF or Rheumatoid Factor positive = Rheumatoid arthritis

ANA positive = Lupus Erythematosus Syndrome (LES)

Normal Values: <500mg/ml FEU or < 250 mg/ml DDU

Nota: can be false if is high in patients with:

- ✓ Liver disease
- ✓ Cancer
- ✓ Myocardial Infarction
- ✓ Pregnant

THYROID

- TSH = 0.2 – 5.4
- T3 = 80 – 230
- T4 = 5-12
- T3 – T4 ↑ TSH↓ = Hyperthyroidism
- T3- T4 ↓ TSH↑ = Hypothyroidism

GI Diagnostic Procedure**1. ERCP (Endoscopic Retrograde Cholangio Pancreatography)**

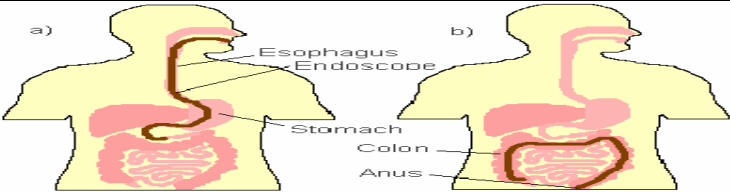
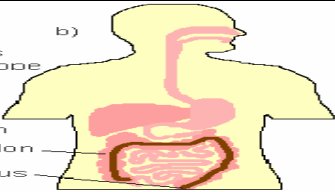
- NPO
- Assess for return GAG Reflex after
- Monitor for Pancreatic Perforation
- Pancreatitis = Board like abdominal pain

2. Occult blood Test

- Guaiac or Hemoccult.
- Sample Collection (caca)

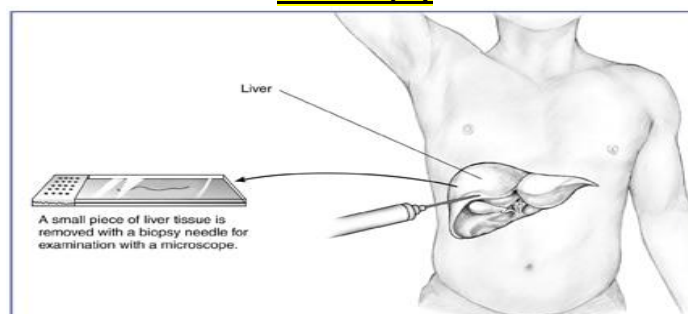
<u>False Positive</u>	<u>False Negative</u>
- Red meat and sausage - Broccoli, turnips, radishes - NSAID - ASA, steroids - Red wine - Bismuto Subsalizilate (peptobismol) Withhold 2-7 days before all those.	<u>Vit C</u> Withhold Vit C 2-7 days before

3. Barium swelling test X – RAYS with Barium

UPPER GI SERIES	LOWER GI SERIES
 Esophagus, stomach, and Duodenum (tragar) No preparation of the intestine Nota: Upper GI studies Test THE PEARLS FOR NCLEX REVIEW COURSE Diagnostic Test: Upper GI Series Facts about the test: • Takes 5-6 hours to complete • Clear liquid diet after supper the evening before the exam • NPO after midnight • Enemas and laxatives are not given in preparation for the procedure • Laxative or enema post-procedure if barium is used • Barium takes 14 days to clear the colon • Gastrografin may cause diarrhea Come visit us to get the full course! BrainyNurses.com Series of X-ray images to follow the contrast through the esophagus, stomach, and duodenum. May be done with barium or gastrografin. Uses: to diagnostic Gerd and hiatal hernia.	 Barium Enema Colon and Rectum Preparation of the intestine needed Barium coats the lining of the colon which shows up on x-ray pictures last part of small intestine caecum rectum anus Barium liquid is poured through thin tube which is passed through the anus Barium Enema
Pregnancy test for both	
NPO Swallowing test before (speech therapy) FEES (fiberoptic endoscopic evaluation swallowing)	Bowel preparation (to clean the intestine) Clear liquid diet Laxative (Polyethylene- glycol- electrolyte solution) NPO Perform before upper GI Perform after ultrasound or colonoscopy
After Procedure	
Plenty fluid Increased fiber Expected white or clay feces for few days after.	Plenty fluid White feces = expected Increased fiber

Estudies GI (en este orden)

1. Ultrasound or Colonoscopy
2. Lower GI
3. Upper GI

Liver Biopsy**Pre – Procedure**

- Coagulation studies (PT, APTT)
- Give Neomycin (to kill bacteria to prevent increased ammonia).
- Hold expiration while the needle is inserted
 - o Position during the biopsy = **Supine or over left side, with right arm under head**
 - o Position after the biopsy over right side support with pillow
 - o Mild pain radiating to right shoulder.

Nota: Si despues de un procedure te duele el lado derecho. Documente.

Los dolores que se corren hacia el lado derecho, no se reportan, solo documente.

Lo unico que se debe reportar es el dolor de un embarazo ectópico lado derecho.

Stomach- Esophagus-Endoscopy

- Informed Consent
- NPO
- After procedure check gag reflex
- Expected some throat for 3-4 days after.

Sigmoidoscopy	Colonoscopy
- Sigmoid colon return an anal canal. - Laxative and Enema or suppository morning of procedure. - NPO	- All colon - Clear liquid diet 24-72 hours - Cathartic in evening 2 days before exam - Glycolytic Solution
If Brown Feces at the morning DO NOT PRACTICE TEST.	

Diagnostic Test Urinary System

Cystoscopy (bladder and Urethra)

- ✓ Bowel Preparation
- ✓ Urine usually pink- tinged after exam = expected
- ✓ Antibiotic prophylactic.

Intravenous Pyelogram

- ✓ Kidney, Ureters, and bladder
- ✓ Bowel preparation
- ✓ NPO
- ✓ Check for iodine or selfish allergies (if metphormin, stop it first and then do the test)
- ✓ X- Ray = Pregnancy Test Before.

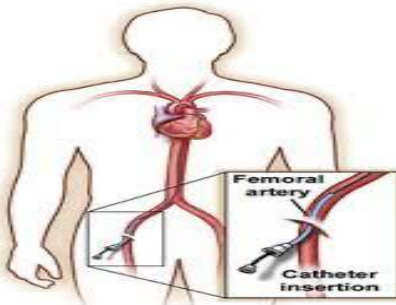
CT Scan

- ✓ Non- invasive procedure
- ✓ Check for pregnancy Test
- ✓ With Constrict – Check for Allergies

MRI

- ✓ Explain Procedure
- ✓ Assess for Claustrophobia
- ✓ Remove all jewelry, metal, patch.
- ✓ No in client with pacemaker or clips
- ✓ If shrapnel (fragmentos de bala) Do not do the test.

Cerebral Angiography

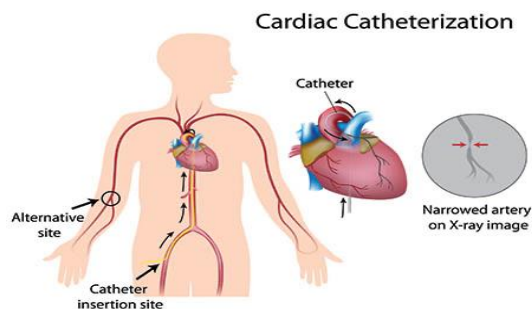


- Informed Consent
- Explain Procedure
- Dye Injection into femoral artery.
- Expected flushing during dye administration.

After Procedure

- Check peripheral pulse (if decreased or absent call Dr.)
- Check for Hemorrhage
- Keep leg in straight position for 8-12 hours after.
- HOB no more than 30°

Cardiac Catheterization

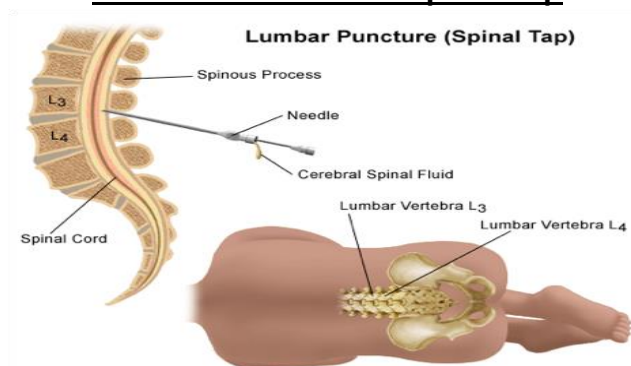


- ✓ History of Allergies
- ✓ NPO (sip of water to take medication is ok)
- ✓ Explain
 - Flushing sensation is expected

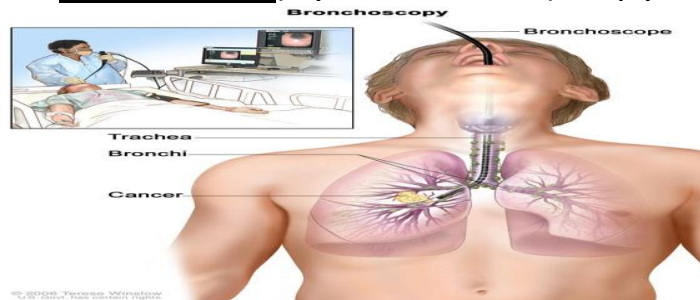
Post procedure

- ✓ Check Distal pulse (if absent call Dr.)
- ✓ Check for chest pain (call Dr.)
- ✓ Pressure Dressing for 3-5 mint.
- ✓ Resting extremity flat 6-8 hours
- ✓ Increased HOB no more than 30°
- ✓ Plenty fluid.

Lumbar Puncture= Spinal Tap



- **Position = Recumbent or fetal position**
- **Informed Consent**
- **After procedure**
 - Supine for 6-8 hours
 - Encourage plenty fluid
 - Headache = expected (give Tylenol)

Bronchoscopy (larynx, trachea, bronchi) + Biopsy

- ✓ Informed Consent
- ✓ NPO
- ✓ Medication Sedation (Diazepam) + Atropine for drooling
- ✓ Remove denture
- ✓ Expected sore throat after
- ✓ NPO until gag reflex return
- ✓ If any respiratory difficulties = all Dr.

Herb Preparation**1- GARLIC**

- ✓ Antibiotic, Antilipemic
- ✓ Antithrombotic, antiinflammatory
- ✓ Decreased BP
- ✓ Is the only one can be given with MAOIs

Side effects

Halitosis.	No in peptic ulcer	Risk for bleeding
Flatulence.	No in reflex	No with anticoagulant
Heartburn	Risk for bleeding	No with antiplatelet
		No with oral Hypoglycemic

Nota: los que empiezan con G = Risk for bleeding

2. Ginseng

- ✓ **Increased BP = No with Maois** (to prevent hypertensive crisis)

Side Effect

- Headache	No with anticoagulant (bc bleeding)
- Insomnia	No with Digoxin (bc ↓ K)
- Palpitation	No with steroids (bc ↓ K)
- Diarrhea	No with steroids
- Vaginal Bleeding	No in Cardiovascular disease (bc ↑ BP)
- Bleeding	

3. Gingko Biloba

- ✓ Enhance cerebral and peripheral blood circulation
- ✓ Use for Dementia
- ✓ Short term memory loss
- ✓ Sexually function

Side Effect Gingko

- **Increased BP** = Headache, Dizziness
- No with MAOIs
- **No with anticoagulant (Risk for bleeding)**
- No with NSAID
- No with ASA

4. Ginger ✓ Antiemetic ✓ Antioxidant ✓ Digestive	Side Effect - No with Anticoagulant (bleeding risk) - No in gallstones (Piedra en riñones)
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5. Milk Thistle

- ✓ Liver Detoxification

6. Hyosiamine

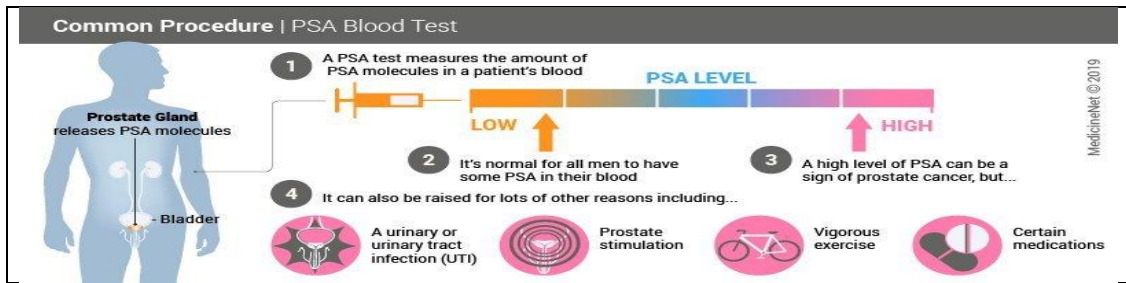
- ✓ Anticholinergic

Anticholinergic cierra todo menos MYDRIASIS  Mydriasis  Bronchodilation  Tachycardia	Anticholinergics Cierra todo lo demás - ↓ Tears - ↓ Salivation - ↓ Urinary retention - ↓ Constipation - ↓ Sweating
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7. Licorice ✓ Expectorant ✓ Antinflammatory ✓ Use for: cough, colds, stomach pain	↓ K = No with Digoxin ↑ BP = No in Cardiovascular disease ↓ Effect Spirinolactone
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8. Fever Few ✓ Antipiretic No with Anticoagulant (risk for bleeding)	Side Effect - Mouth Ulcer - Heartburn - Indigestion
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9. Saw Palmetto - Mild diuretic - Urinary Antiseptic - Used for BPH (Benign prostatic hyperplasia) - Increased sexual vigor.	Side effect - Constipation - Back pain - Decreased Libido May cause false Negative of PSA test, withhold 72 hours before PSA
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**10. Echinacea**

- Immunostimulant
- Antiviral
- Antibacterial

Side Effect and Contraindication

- No in Autoimmune disease
- LES
- CHRON
- COLITIS
- RA
- ↓ Effect Immunosuppressant
- Unpleasant taste on tongue.

11. Evening Primrose oil

- Antinflammatory
- Sedative
- Use for Pre- menstrual and post- menopausal problem

Side Effect

- Seizure no in epilepsy or schizophrenia.

12. Chondroitin Glucosamine

- Use for Arthritis and Osteoarthritis

Side Effect

- Increased blood glucose level.
- No in diabetic.

13. Bilberry

- Decreased Intraocular pressure
- Use for Cataracts
- Glaucoma
- Diabetic Retinopathy
- Night blindness
- Use for Diabetes (decreased blood glucose)
- No with Disulfiram (quitar el deseo de alcohol)

14. Anxiety Herbs**1. Chamonile**

- Sedative
- Antinflammatory
- Antispasmodic
- Use for GI disorder
- Insomnia

No with

Antianxiety herb or medication.

- Benzo
- Buspirone

Nota: SSRI = Antidepressive

Buspirone = anxiety, stress

Busproprium= Depression											
<u>Buspirone opposite busproprium</u>											
2. <u>KAVA</u> - Antianxiety - Sedative, hypnotic	<u>Side Effect</u> - Liver toxicity - Suicidal thoughts - Visual problem <u>Contraindication in:</u> Parkison Stroke Depression										
3. <u>Melatonin</u> - Use for Insomnia - Use for jet lag  -Do no use with: ✓ Kava ✓ Diazepam ✓ Buspirone	<u>Side Effect.</u> Headache Confusion Tachycardia										
4. <u>Valerian</u> Sedative, hypnotic	<u>Side Effect</u> Headache No in liver diseases No with antianxiety No with Benadryl										
15. <u>Antidepressive</u> - St- John's wort - Use for mild to moderate Depression	<u>Side Effect</u> - No in major depression - No with MAOIS, SSRI or Tricyclic - Wait 2 weeks between - No with Tyramine containing										
<table><tr><th>Foods to Avoid That Contain Tyramine</th></tr><tr><td>Avocados</td></tr><tr><td>Bananas</td></tr><tr><td>Beef or chicken liver</td></tr><tr><td>Brewer's yeast</td></tr><tr><td>Broad beans</td></tr><tr><td>Caffeine, such as in coffee, tea, or chocolate</td></tr><tr><td>Cheese, especially aged, except cottage cheese</td></tr><tr><td>Meat extracts and tenderizers</td></tr><tr><td>Overripe fruit</td></tr></table>		Foods to Avoid That Contain Tyramine	Avocados	Bananas	Beef or chicken liver	Brewer's yeast	Broad beans	Caffeine, such as in coffee, tea, or chocolate	Cheese, especially aged, except cottage cheese	Meat extracts and tenderizers	Overripe fruit
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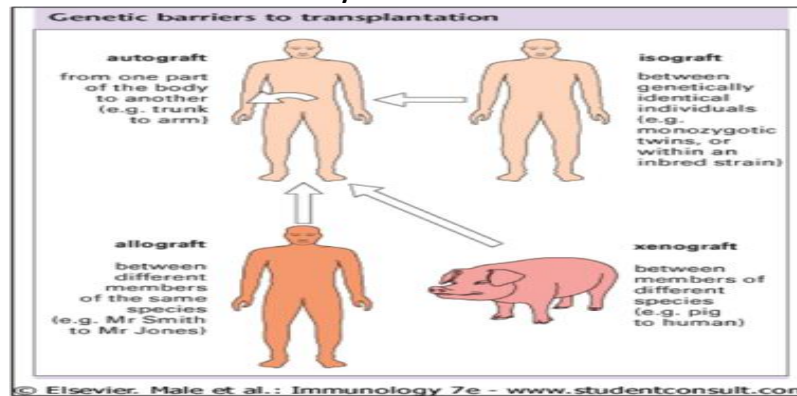
16. Black Cohosh

- Use for postmenopausal symptoms. (Hot flushes, vaginal dryness)
- Use to treat side Effect of TAMOXIFENE

Nota: Tamoxifene = use cancer de mama
= Side effect DVT.

1. Integumentary System**➤ Autograft Position**

- Site immobilized for 3-7 days.

**➤ Face Burn**

- Semifowler to fowler position
- Circunferencial
- Leg or extremity elevated above the heart level.

2. Reproductive System**- Lithotomy Position**

- **Labor** = At the end of transition or at the beginning of the 2nd stage (10 cm of dilation)

➤ Mastectomy

- Semifowler position with the affected arm elevated above the heart level.

Nota: No se cortan las uñas en ese lado, tiene que ser el podólogo.

➤ Supine Hypotension

- Side Lateral (Left)

➤ Late deceleration

- Side lateral

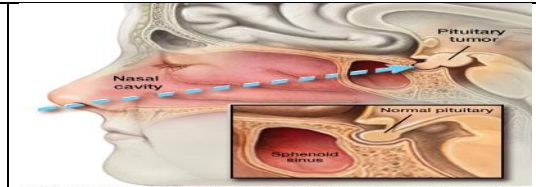
➤ Variable deceleration

- Knee Chest
- Trendelemburg
- Recumbert
- Side Lateral

- **Risk for Pre-Term labor**
- Trendelenburg over left side.

3. Endocrine System

1. Hypophysectomy = HOB 30°



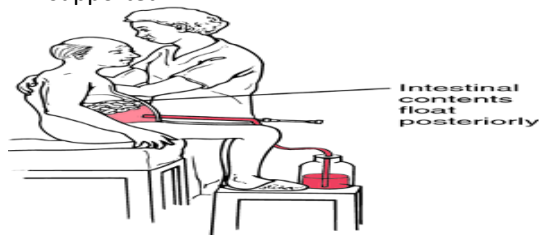
2. Thyroidectomy semifowler to fowler

- Support head with sand bag
- Or place your hand under head to prevent neck hyperextension.

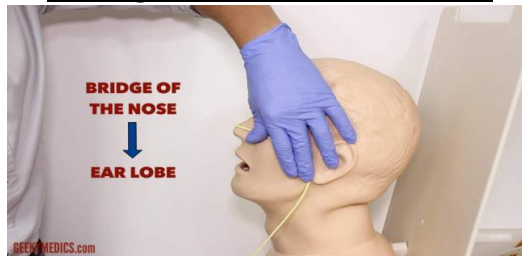


4. GI System

- **Hemorrhoidectomy** = Lateral side lying.
- **GERD** = Reverse Trendelenburg or elevate HOB with 6–8-inch block during night.
- **Liver Biopsy**
- Supine or left with right arm under neck.
- After = Right side
- **Paracentesis**
- Semifowler position in bed or upright on the side of the bed or in chair with feet supported.



Nasogastric Tube Insertion



- High fowler's position 90°.
- With head tilted forward (sip of water to help going down)

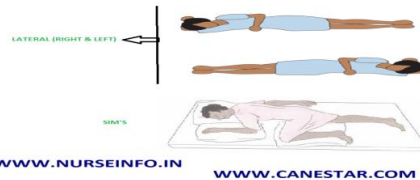
Irrigation and tube Feeding- Semifowler to Fowler

Intermittent
30° - 45° HOB
1 - 2 hrs After

Continuous
All Time 30° - 45° HOB
1 hours supine for client care.

Rectal Enema

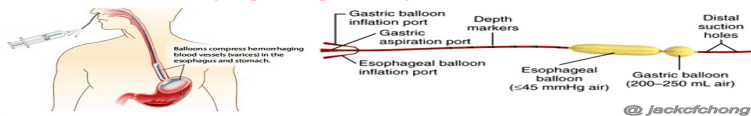
- Left Sims Position
- Lateral Position

PATIENT POSITIONING - LATERAL & SIM'S POSITIONSengstaken - Blakemore Tube**EMNote**

Sengstaken-Blakemore Tube

To stop UGI bleed from esophageal varices (EV)

- Routine **ETT intubation** before SB tube
- Inflate gastric balloon in 2 steps: 50cc → **X-ray** → 200-250cc
- **1-kg traction**: gastric balloon compress GE junction → reduce EV blood flow
- Inflate esophageal balloon (**< 45mmHg**) if traction failed to stop bleeding
- Esophageal balloon should not remain inflated **> 6 hours** (avoid necrosis)
- Admit to ICU: **esophageal or gastric ruptures** are not uncommon



- Use for esophageal varices
- Intermittent
- 30° - 45° HOB
- Scissors at bed side to cut gastric balloon in case of respiratory compromised.

Board: Scissor = Sengstaken – Blakemore
 Wrench = Halo Fixation devices
 Wirecut = Jaw Fracture

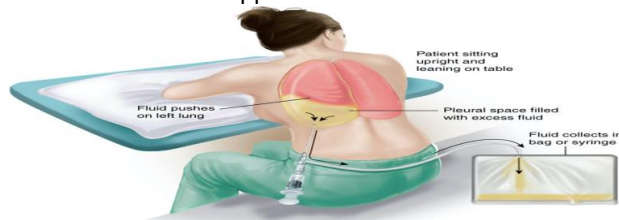
Respiratory System

➤ Semifowler to Fowler position

- COPD = 30° - 45°
- Laryngectomy = 30° - 45°
- Bronchoscopy = 30° - 45°

➤ Thoracentesis

- Sitting on the edge of the bed or examining table and leaning over the bed side table with feet supported on a stool.



➤ Pneumectomy or lobectomy

- Over affected side.

➤ Pneumonia

- Over unaffected side.

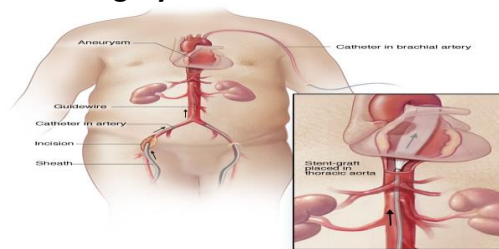
Nota: Si quiero que drene, lado opuesto, si no quiero que drene, mismo lado.

➤ ARDS. Fat Embolism = Prone Position

Cardiovascular System

1. Abdominal Aneurysm

- After surgery HOB no more than 45°



2. Amputation of Lower extremities

Above the Knee or below the Knee

<u>During first 24 hours</u>	<u>After 24 hours</u>
- Elevate residual limb or stump on a small pillow to decreased Edema. - BKA = Increased Foot of bed	- Prone position alternate with side lateral for 20 – 30 minutes 3 – 4 time x day to prevent flexion contracture

3. Arterial Vascular Graft

- Affected extremity kept straight for 24 hours.

4. Heart Failure = Fowler to high fowler

- Acute Pulmonary Edema } first intervention
- Pink frothy sputum } Sitting position

Peripheral Arterial Disease	DVT, Varicose vein
Legs dangling or in dependent position	Leg Elevated

Arterial vs. Venous Lesions (location & appearance)?	
Arterial system  Location • End of toes • Top of feet (dorsum) • Lateral ankle region (malleolus) Ulcer's Appearance? • Very little drainage • Little tissue granulation (pale/very light pink) or (necrotic/black) • Deep "punched out" w/ noticeable margins/edges that gives it a neat appearance.	Venous system  Location • Medial parts of lower legs • Medial ankle region Ulcer's Appearance? • Swollen w/ drainage • Granulation present (deep pink to red) • Edges irregular • Shallow

Sensory System

➤ Cataracts Surgery

- Semifowler to fowler over unaffected side.
- Place personal belongings close to unaffected side



➤ Retinal Detachment

- Over the detached side

Nota: The diabeters rethinopathy can lead to a retinal detachment.

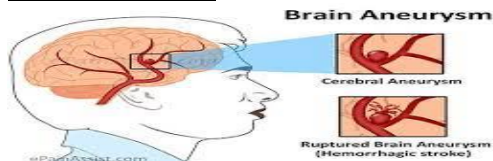


Neurologic System HOB = $\uparrow 30^\circ$

1. Autonomic Dysreflexia = sitting position

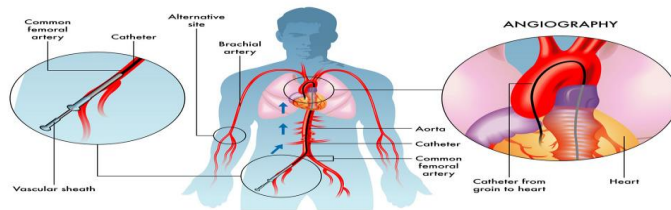
1. _Sitting
2. Check bladder distention
3. Check Constipation

2. Cerebral Aneurism = $30^\circ - 45^\circ$ HOB



3. Cerebral Angiography

- Extremity kept straight and immobilized for about 6 -8 hours.

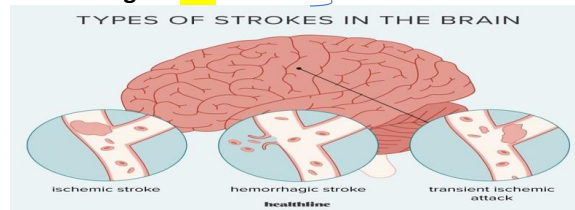


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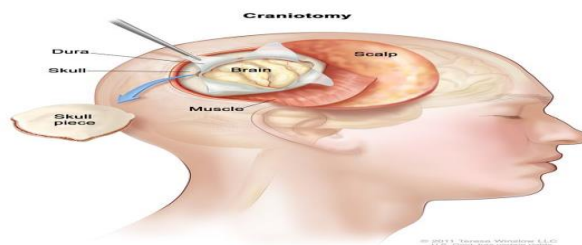
4. STROKE

- Ischemic = Flat
 - Hemorrhagic = $30^\circ - 45^\circ$
- head at the midline



5. Craniotomy

- ICP = $30^\circ - 45^\circ$
- No more than 45° or less than 30° bc = can increase ICP
- Midline

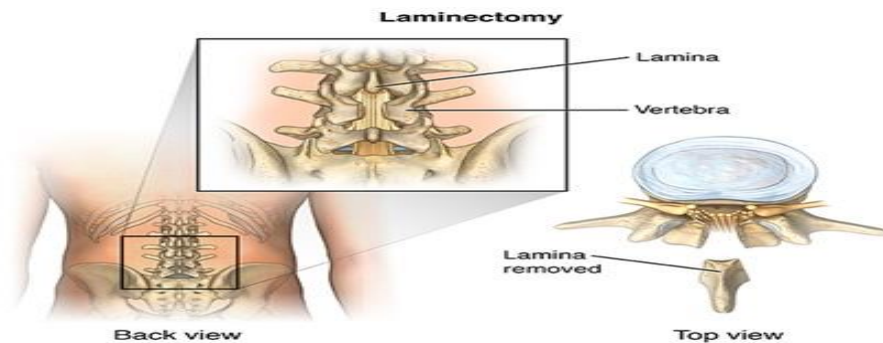


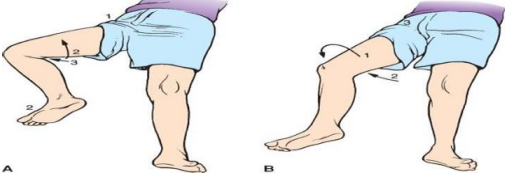
6. Ventricular peritoneal Shunt

- Supine over unaffected side

**7. Laminectomy or other vertebral surgery**

- Back Straight = Log rolling technique to change position (2 persons)

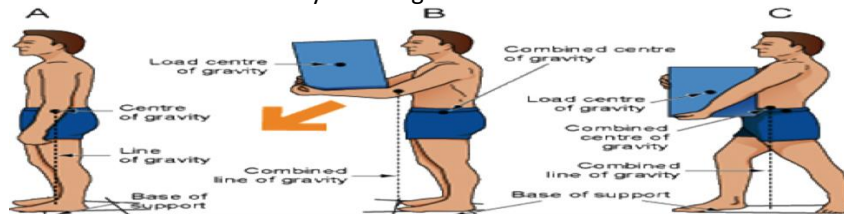
**Musculo-skeletal****➤ Total hip replacement**

Hip Fracture	Hip Replacement
- Leg Shortening - External rotation - Adduction - Limited Abduction  Nota: el board los niños tienen esa position because developmental displasia hip	- Alignment = use a trochanter roll. - <u>Abduction</u> = use wedge, triangular or abductor pillow  Hip Abduction Pillow

- Sitting Position = 90° Feet on the floor
- **On bed = HOB no more than 60 (even if risk for aspiration)**

Body Mechanism (Ergonomic Principle)

- Encourage the client to assist as much as possible.
- Keep the back, neck and pelvis aligned (avoid twisting).
- Flex knee and keep feet wide apart.
- Rise the client bed so that the client weight is at the level of the nurse center of gravity.
- Position Self close to the client.
- Use arms and legs (no back).
- Slide client toward yourself using a pull sheet.
- The person with the heaviest load.
- The team involved by counting to 3.



Vitamins

- FAT Soluble Vitamins = A, D, E, K

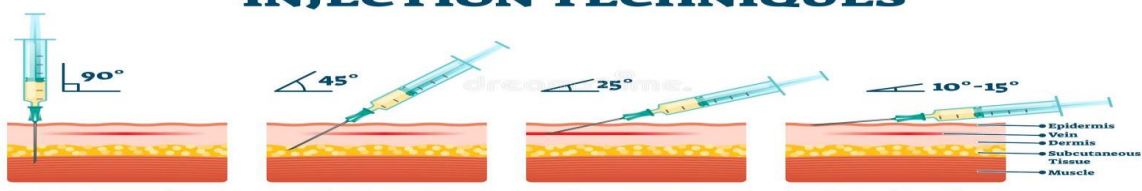
Vits.	Function	Primary Source	Clinical Manifestation
A	➤ Visual Acuity. ➤ Adaptation to light and dark. TTO for Acne. Isotretinoin or Retinoic A - No tomar con nada amarillo.	B – Carotene - Carrot - Yam - Pumpkins - Summer Squash - Sweet potatoes - Eggs yolk - Butter - Cheese - Buttermilk - Peaches ➤ Vit A Vit A derivate = Avoid all these food	- Deficiency night blindness. <u>Toxicity</u> CNS Change Portal Hypertension
D	➤ Bone Calcification	- Fish oils - Fortified milk dairy - Sunlight Irradiation	- <u>Deficiency</u> = Rickets child Osteomalacia Adult. Weak bone(se doblan). <u>Treatment</u> Solo Vit D pie sambo o bow legs. Dental hypoplasia. <u>Toxicity:</u> Hypercalcemia Renal Calculo.

E	➤ Antioxidant ➤ Growth	- Liver - Green Leafy vegetable. - Oil - FATS - Nuts - Grain	- Deficiency Hemolytic Anemia Fatigue Headache Blurred vision Diarrhea
K Phyto nadio ne	➤ Warfarin antidote ➤ Coagulation ➤ Synthesized by intestinal bacteria. - Administered all newborn IM = vastus lateral withing 72 hours	- Green leafy - Vegetables - Eggs - Cheese	- Deficiency Bleeding Toxicity - Anemia - Thrombosis - Liver and Renal Damage. - Intestinal Bacterial Infection
WATER SOLUBLE VITAMINS			
B1 Thya mine	➤ Normal Growth ➤ Carbohydrate metabolism Toxicity - Shock. - Nota: PT chronic alcoholism: 1- Wernicke encephalopathy 2- Korsakoff Psychosis	- Legume - Meat - Enriched grain - Eggs - Fish Treatment - Thymine = Vit B1 (pills)	- Deficiency Beri beri: ✓ Polyneuropathy (numbness, Tingling) ✓ Fatigue ✓ Decrease DTR reflex ✓ Muscles wasting.
B3 Niaci n	➤ Normal Growth Nota: Nicotinic Acid = Tto to decrease cholesterol (es la misma vit B3 con otro nombre).	- Enriched grains - Meats Board: PT with nicotinic Acid administered aspirin or ibuprofen to prevent flushing. Best option = Aspirin	- Deficiency ✓ Pelagra: 3Ds  VITAMINA B3 (NIACINA) El ácido nicotínico se convierte con facilidad a nicotinamida. Es esencial en forma de coenzimas NAD y NADP. Forma activa: NAD. El Ag. es el Nicotínico. El Ag. produce liberación de energía. La PELLAGRA puede presentarse subclínicamente como: La deficiencia de nicotina se caracteriza por: alteraciones dermatológicas y mentales. 1. Dementia 2. Diarrhea 3. Dermatitis (Circumferential broad collar rash) ✓ Glossitis ✓ Decrease weight ✓ Rough scaly skin

B6 Pyridoxine	➤ Amino Acids Metabolism	- Meat - Legume - Corn - Soy - Liver - Eggs yolk - Sunflower	- Deficiency 1. Peripheral Neuropathy Administer Vit B6 with Ionized to prevent peripheral Neuropathy 2. Anemia 3. CNS change (seizure)
B12 Cobalamin	➤ RBC Formation	- Meat - Organ Meat (liver and Kidney)	- Deficiency Pernicious Anemia + Red Beefy Tongue + Neurology Symptoms Treatment B12 Long life IM
C Ascorbic Acid	➤ Antioxidante ➤ Healing	- Orange and orange juice. - Tomatoes - Guava - Broccoli - Spinach - Cabbage	- Deficiency ✓ Scurvy Receding gums that are spongy prone to bleeding (se caen los dientes) ✓ Anemia ✓ Bone pain ✓ Joint pain ✓ Petechiae ✓ Swelling edema ✓ Corks crew hair (El pelo en candela) ✓ Poor wound healing

Malnutrition types	
KWASHIORKOR	MARASMUS
- Edema - Anemia - Fatty Liver - Edema and swelling belly	- Muscle wasting - Loss of subcutaneous FAT - Skinny to very skinny
	

IV DEVICES

1. BUTTERFLY SETS - Use for small vein (children and elderly) - Risk for infiltration	
2. Needle - Gauge (16-26) inverse + gauge = + smaller - Length = 0.5 Subcutaneous - = 1.5 inch= Intramuscular	
Nota: Insulin administration * Subcutaneous $\nless 45^\circ$ - If overweight or obese, BMV > 25, administer $\nless 90^\circ$ - Gauge 25-26 - Length 0.5 – 0.6 – 5/8 Do not massage	Nota: Heparin Administration * Subcutaneous $\nless 90^\circ$ - If underweight or malnutrition (cacketic), Administer $\nless 45^\circ$. - Gauge 25-26 - Length 0.5 – 0.6 – 5/8 - Do not massage They are opposite.
INJECTION TECHNIQUES 	

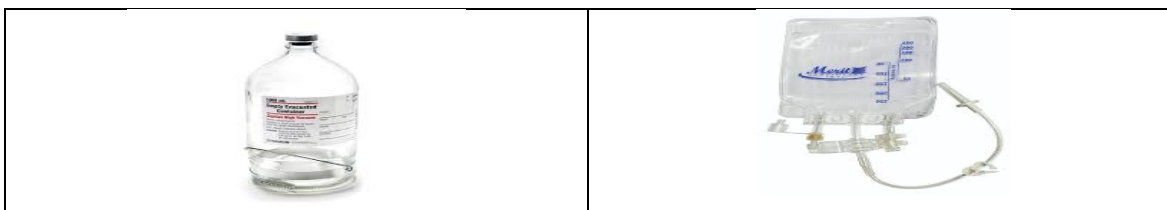
IV GAUGES

1. Rapid emergency = 16-18
2. Blood transfusion= 18-20
3. FAT peripheral infusion= 20-22
4. Small veins = 24-26 (also subcutaneous)
5. Standard IV fluids = 22-24

IV Cannula Gauge Sizes & Color Code -VISUAL MNEMONIC			
COLOR	GAUGE	Flow Rate	Recommended Use
VIOLET	26	13mL/min	Elderly & Neonates
YELLOW	24	20mL/min	Very Fragile veins, Elderly & Pediatric patient
BLUE	22	36mL/min	Chemotherapy infusions, elderly and pediatric
PINK	20	60mL/min	* Multipurpose IV * For medications & Hydration
GREEN	18	90mL/min	Blood Transfusion, Large volume infusion
GREY	16	180mL/min	Trauma, Surgery, Large volume infusion
ORANGE	14	240mL/min	Massive trauma

IV CONTAINERS

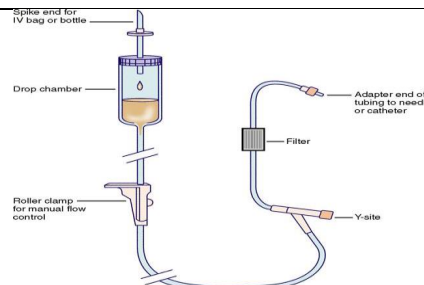
Glass ✓ Check for any cracks before hanging	Plastic ✓ Squeeze to ensure intactness. ✓ Do not write with marking pen to prevent ink absorption. ✓ Use a ball point pen or label.
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IV Tubing

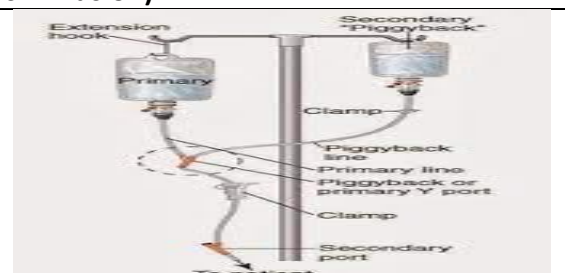
- ✓ Spike
- ✓ Drip Chamber
- ✓ Roller Clamp = **Open and prime the line before hanging.**
- ✓ Y – site.
- ✓ Adapter = connected to the IV line.

Nota: Extension tubing can be added and IV Tubing set to provide extra length for children and Pt who have special mobility needs.



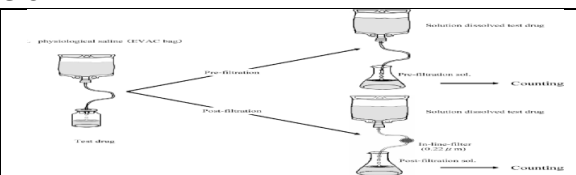
IV PB (IV piggyback Infusion)

- **To administer intermittent infusion without disconnecting existing IV:**
 - ✓ Set up secondary set
 - ✓ Hang existing infusion set (primary).
 - ✓ Check for compatibility between bags before hanging.
 - ✓ Cleanse upper infusion port with alcohol (dry for 30 seconds).



Filters

- **0.22 μ m** is used for most solution.
- **1.2 μ m** is used for **lipid and albumin.**
- **Change filters every 24-72 hours**



Electronic IV Infusion Devices

IV Infusion Pump

- Control amount of fluids.
- Ex: K Cl, parenteral nutrition

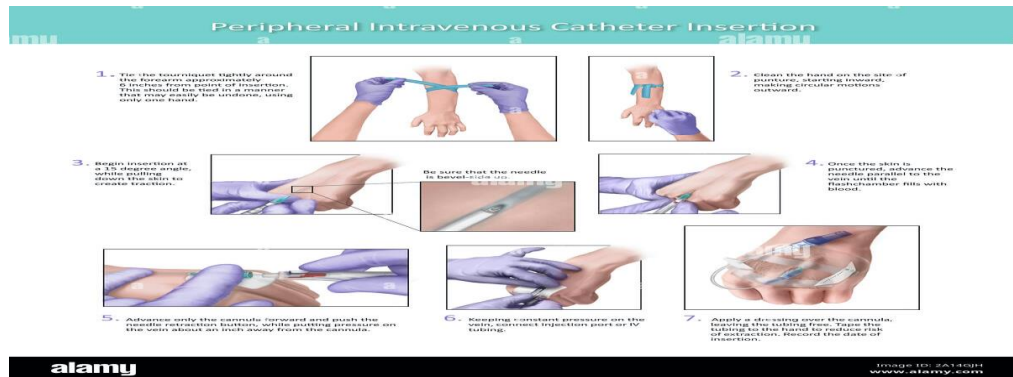
INFUSION PUMP



Patient Control Analgesic PCA

Ex. Pt has to be oriented x 3





Selection of the peripheral IV Site

- Vein in the hand, forearm and antecubital fossa are suitable for adult.
 - Veins in the lower extremities L/E (legs and feet) and scalp are suitable for infant.
 - Start the IV infusion distally in non-dominant hand.
 - Do not place restraint over venipuncture site.
 - Avoid sites over joints.
 - **No in paralyzed arm**
 - Avoid BP check in IV site
 - Used arm board when venipuncture site is localized near flexion areas (specially children).
 - No over IV shunt (fistula) or lymphedema.
 - No over site with dermatitis
- A

B

C
- No in mastectomy.

Steps for IV Line Administration

1. Lower the arm to visualize vein.
 2. Place tourniquets 3-4 inches above insertion site.
 3. Prepare skin.
- Clean with alcohol or chlorhexidine wipe in circular motion (inner to outer).
 - Allow dry for 2 minutes.
4. Introduce needle 10°-30° with bevel up hold needle still and advance catheter.
 5. After blood return, release tourniquets.
 6. Connect catheter to IV tubing and apply sterile dressing to insertion site. Label date and initial.
 7. Initiates prescribe flow rate.
 8. Dispose sharp.
 9. Document

- Change venipuncture site every 48-72 hours.
- IV dressing every 72 hours.
- IV Tubing change
- IV bag open 24 hours.
- Before adding medication to solution swab access port with 70 % alcohol.

IV Complications

1. Pulmonary embolism (air embolism).

✓ To prevent

- Prime the line.
- Secure all connection
- Teach patient to bear down or withhold respiration during insertion and removal.

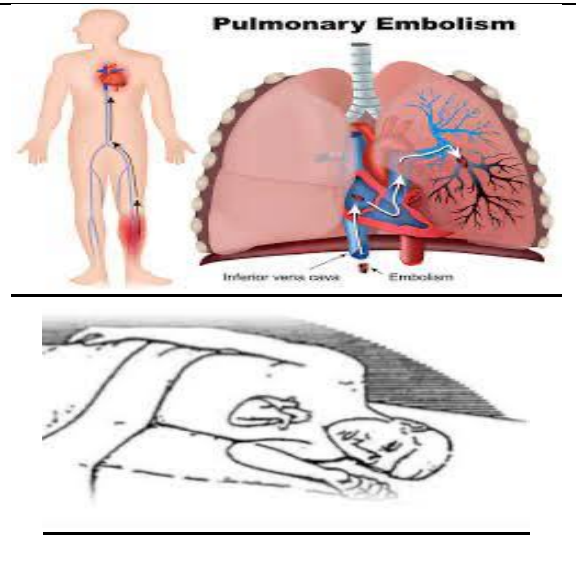
✓ Central venous catheter position

- Administer in supine or Trendelenburg position with head away to insertion site.

Nota: Trendelenburg position if not head injury.

✓ If suspected Air embolism

1. Clamp the catheter.
2. Trendelenburg over left side
3. O₂
4. Call the Dr.



2. Hematoma

- Collection of blood in the tissue after unsuccessful venipuncture or after discontinued IV line.

✓ Prevention and intervention

- When starting an IV line avoid piercing the posterior wall of the vein.
- Do not apply tourniquet to the extremity immediately after unsuccessful venipuncture
- When discontinued an IV, apply pressure for 2-3 mint and elevate extremities.
- If Hematoma develop, elevated the extremity and apply ICE.



3. Infiltration

- Is a form of tissue damage of the IV fluid out of vein into surrounding interstitial spaces

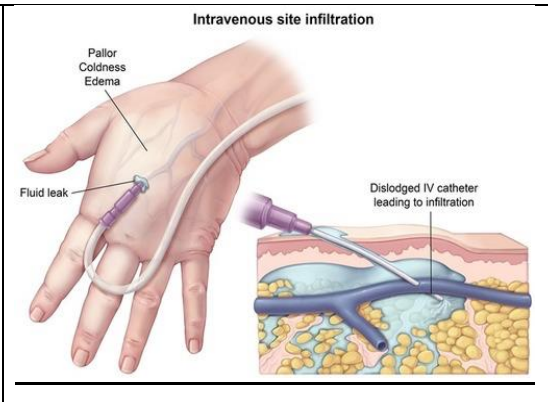
Assesment

- Swelling
- Palor
- Coldness

Board: Como sabes si hay infiltration?

1. Ocluye o tapa la vena
2. Baja la bolsa mas abajo

Si hay sangre en la bolsa todo esta bien, no infiltration

**Prevention- Intervention for Infiltration**

1. Avoid venipuncture in flexion areas.
2. Monitoring IV rate for decreased or stop of flow.

Evaluate the IV site to check for Infiltration

- Occluding the vein proximal to the IV site
 - o If the infiltration continues to flow = infiltration
 - o If stop = IV still in vein
- Lower IV fluid container and check for blood return, if blood in bag = IV still in vein.
- 3. Stop IV can be restart proximal to the infiltration**
 - Elevated the extremity
 - Warm compress

4. Extravasation

- Swelling
 - Redness
 - **Blister**
 - Necrosis
 - Infiltration of vesical (chemotherapy)
- Or vasoconstrictor (dopamine, epinephrine)

**Intervention for Extravasation**

1. Stop IV immediately.
2. Removing tubing, IV bag and maintain the line
 - ✓ Introduce a syringe and remove 5-10ml of blood
3. Administer Antidote subcutaneous
 - ✓ Chemotherapy = Hyaluronidase
 - ✓ Vasoconstrictor = Phentolamine (emergency bc decreased BP potent)
4. Apply Ice pack

5. Phlebitis

- Vein inflammation.

Assessment

- Redness
- Hot

Thrombophlebitis (superficial thrombosis)

- Cord formation or streak through vein wall.



Intervention & Precaution

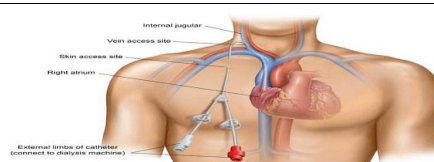
- Avoid using L/E in adult
- Avoid Flexion Areas
- Change venipuncture site every 48-72 hours

If phlebitis occurs

- Stop IV remove
- Notify HCP
- Apply moist warm compress
- Re-start in the opposite site

* Central venous Catheter

- Is used to delivered hyperosmolar solution.
- To measure CVP, infuse TPN, for infusion of different medication for hemodialysis.



Types of CV Catheter

<u>Tunneled</u>	<u>Non-tunneled</u>
✓ More Permanent ✓ Use for long term medication - Hickman - Broviac - Groshong	✓ Temporally (superficial) ✓ Use for short Term. ✓ More risk for infection ✓ Hemodialysis (while fistula is ready)

Catheter position is **determinate by X-ray**

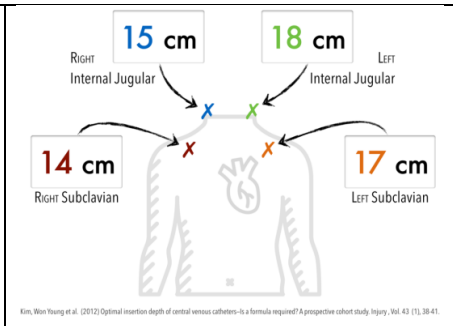
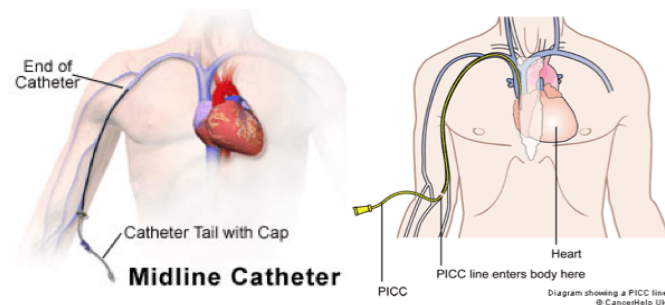
■ Catheter Inserted

- **Internal** jugular or **subclavian** vein.
- Or peripherally insertion through the **basilic or cephalic** vein of the arm
- PICC line = In place + 6 weeks.
- Mid line = in place 2-4 weeks

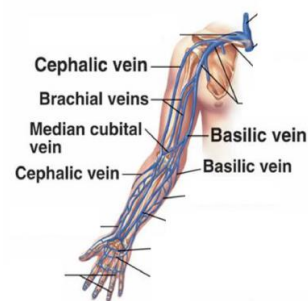
✓ **CVC = hospital**

Home

- PICC = more than 6 weeks
- Mid = 3 – 4 weeks
- Peripheral = 72 hours



Kim, Won Young et al. (2012) Optimal insertion depth of central venous catheters—is a formula required? A prospective cohort study. Injury, Vol. 43 (1), 38-41.



Multi-lumen Catheter

- More than one medication can be administered at the same time without incompatibility problems



Complication Central Lines

1. **Infection** (do not disturb dressing)
2. **Pneumothorax**
- **Skin crepitus**
- Decreased or absent unilateral breath sound
3. **Pulmonary embolism** = Hemoptysis, tachypnea, SOB, Chest pain.
4. **Phlebitis**
5. **Infiltration**

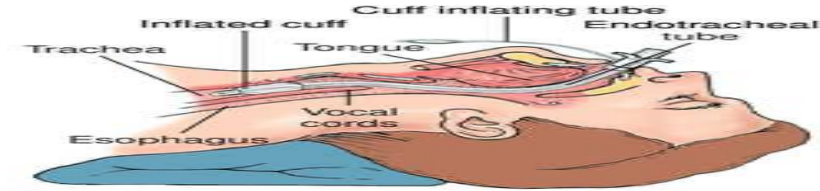
Catheter Care and flushing (RN)

- Flushing solution NaCl 0.9% or Heparin.
- Syringe use for flushing > 10ml
- Using aseptic technique 5cm or 2 inches area around with alcohol swap in circular motion. (Inner to outer)

Board: No flushing with 5ml or less.

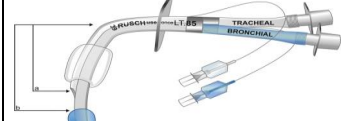
Respiratory System Tube

1. Endotracheal Tube



- Use to maintain a patency airway.
- Indicated when patient needs mechanical ventilation
- In place for 2 weeks
- Cuff located at the distal end of tube inflated 20 mmhg to prevent aspiration.
- In case of mechanical ventilation must be inflated all the time.
- Placement is confirmed by X-ray. (1-2 cm above carina, never aspirated below carina bc perforation).
- Perform auscultation over the stomach to rule out esophageal intubation.
- Monitor the position of the tube at the lip or nose.

Board: Endotracheal tube 2, 3cm out = Call Dr follow up.

<u>Tracheostomy (obturator at bed site)</u>		
Single lumen	Double lumen	Fenestrated tube
✓ Outer canula ✓ Obturator	✓ Outer canula ✓ Inner cannula ✓ Obturator	✓ Pre-Cut fenestrated in the upper posterior wall of the outer cannula. ✓ Allows patient to speak
		

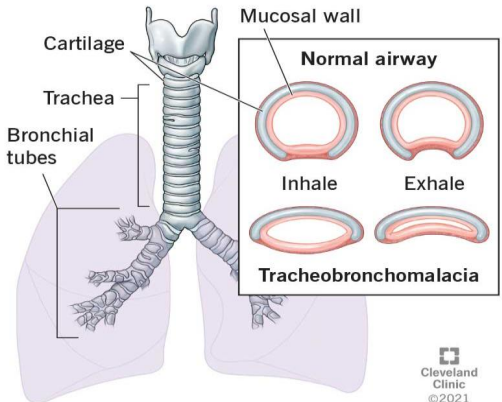
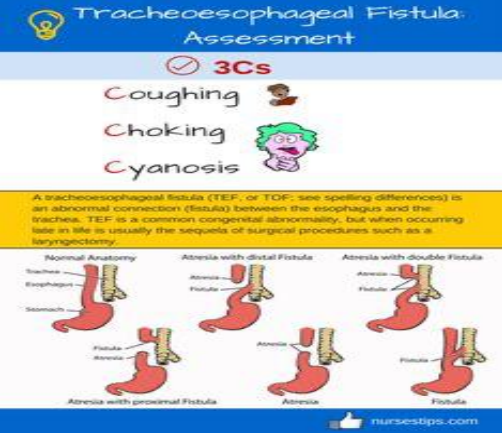
Tracheostomy Care (LPN)

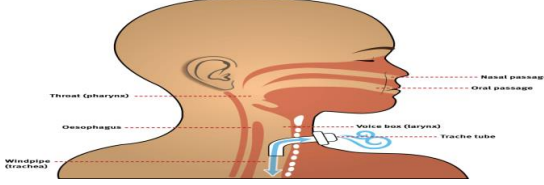
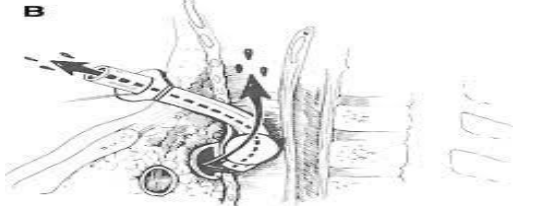
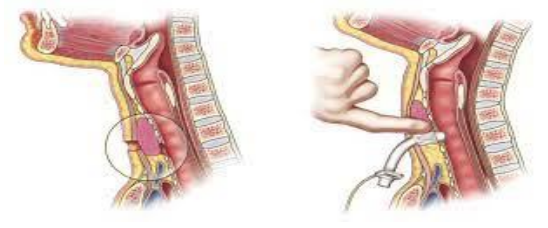
1. Explain the procedure.
 2. Hyperoxygenated 100% O₂.
 3. Suction tracheostomy.
 4. Remove old dressing (use clean gloves).
 5. Open sterile T care kit.
 6. Put on sterile gloves (both hands)
 7. Remove inner cannula, clean with sterile water and dry.
 8. Re-insert into outer cannula.
 9. Clean the stoma with hydrogen peroxide, rinse sterile water and dry.
 10. Change ties or Velcro tracheostomy tube holders as needed.
 - 2 nurses = old tie must remain in place until necktie is secure on side the neck, allow 1-2 finger inserted under tie.
 11. Apply new sterile dressing (pre-cut 4x4 gauze)
 - Never use cotton balls.
 - Never cut gauze in front of client.
- Board:** que lleva follow up = nurse cut gauze in front of client.
12. Documents site of tracheostomy, type and quality of secretion.

Tracheostomy Suction

1. As needed.
2. Position HOB 30°- 45°.
3. Wall suction pressure
 - 100 – 120 mmhg adult
 - 80-100 mmhg children.
4. Prepare equipment, pour normal saline in a basin before doing sterile glove.
5. Keep dominant hand sterile
6. Hyper oxygenated 100 % O₂
7. Test and lubricate catheter.
8. Insert catheter without suctioning until mild resistance (cough and gagging)
9. Remove catheter 1 cm and start suction
 - Intermittent suctioning with circular motion during withdrawal.
 - Limit suction time
 - ✓ 10 s for adult
 - ✓ 5 s for children.
10. Reassess client.
11. Suction nonsterile area (mouth)
12. Continue tracheostomy care.

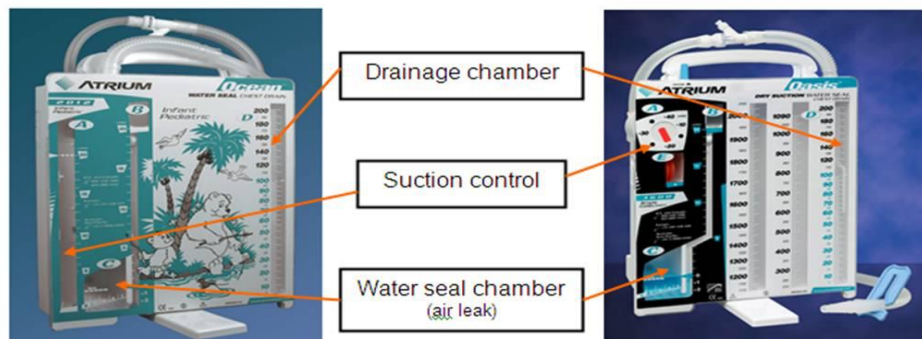
Tracheostomy complication

1. Tracheomalacia	2. Tracheoesophageal Fistula
- Constant pressure exerted by the cuff cause tracheal dilation is erosion of cartilage. Food particles are seen in airway secretions. - No special intervention unless airway problem occurs. - Continue observation.	- Communication between trachea and esophagus (hole). - Food particles are seen in airway secretion. - Increased Coughing - Increased choking while eating Intervention Gastrostomy (PEG TUBE) Jejunostomy (JEG TUBE)
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3. Tracheal stenosis -Narrowed of tracheal lumen Assessment - Coughing - Inability to expectorate secretion - Difficulty breathing and talking Intervention - Tracheal dilation or surgical Intervention	4. Tracheal Innominate Artery Fistula (Red Code C) Assessment - Tracheostomy tube pulsate in synchrony with heart beat. - Heavy bleeding through stoma - Signs of hypovolemic shock. Intervention - Remove the tracheostomy tube immediately. - Apply direct pressure and call Dr.
	
5. Tracheostomy dislodgment (Life threatening emergency) Assessment - Restlessness - Breathing difficulty - Wheezing stridor - Noise Respiration	
INTERVENTION	
During first 72 hours - Manually Ventilation with Ambu bag - Another Nurse call the Rapid Response Team. (bc the stoma is not mature)	After 72 hours - Extend the client's neck. - Grab the retention suture. - Use tracheal dilator or curve forced or clamp to keep stoma open. - Reintroduce obturator and reinsert tracheostomy tube.

Chest Tube

Chest drain system



Suction Control	Water Seal Chamber		Drainage Collection
- Continuous gently bubbling. - If vigorous or intermittent, Call Dr. - - 20 cm water.	- Intermittent bubbling.	- Tiding or fluctuation (como el mar) - Up inspiration - Down expiration - if stop tiding of the tubing : 1. KINK 2. Lungs re-expansion = getting better – auscultation.	- More than 100nl of drainage = Call Dr. - 1st 24 hours expected 500-600ml. Board: must mention the 24 hours - 70 – 100 every 4 hours. - Call Dr. if bright red drainage.

1. Nasogastric Tube Insertion

- Explain the procedure. -
- Position 90° client in high fowler sitting position.
- Determine which nostril is more patent.
- Measure the length of tube front the bridge of the nose to the earlobe to the xiphoid process. Mark with a piece of tape.
- Lubricate tip with water soluble lubricant.
- Gently insert the tube chin down or tuck chin or take water during procedure.
 ✓ If resistance is met (gagging, cough) = Remove the tube.
- Following insertion – X-ray to confirm placement.
- Connect the tube to the suction
- Secure tube to the patient nose with tape.
- Check residual volumen = Most accurate method to confirm placement.
- Check residual volumen every 4 hours.
 - ✓ Aspirate gastric content.
 - ✓ Ph = 3.5- 4 or < 3.5 = ideal
 - ✓ Less than 100ml = ok reintroduce.
 - ✓ More than 100ml = withhold feeding, discard and check in 1 hours.
 - ✓ More than 200 ml = Stop feeding and call Dr.

Other gastrointestinal tubes

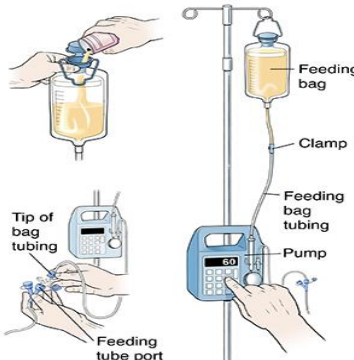
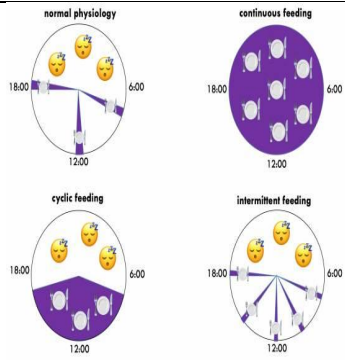
2. Gastrostomy tube (PEG)	3. Jejunostomy Tube (JEG)



Tube Feeding (LPN)

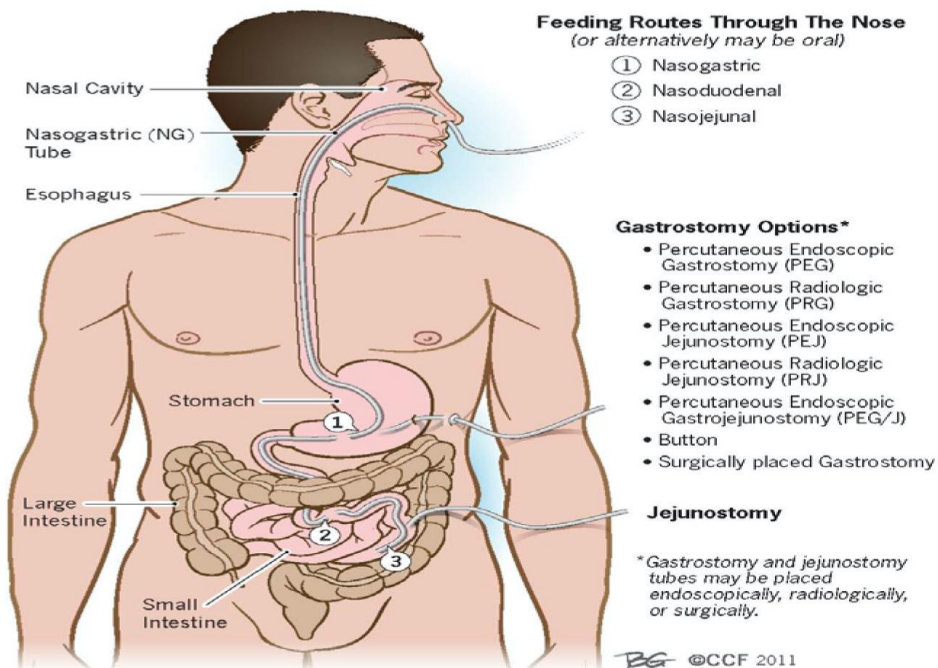


Types

Bolus	Continuous	Cyclical or intermittent
Formula is administered over 30-60 mint, period every 3 – 6 hours via gravity= syringe empty itself.	- Infusion pump is administered through 24hours. - HOB -30° – 45° all the time. - 1 hour flat for client care	- Infusion pump 8-16 hours. - HOB – 30°-45° hours before and 1-2 hours after feeding.
		 Comparison of various feeding schedules, within the context of circadian rhythm.

Administration of feeding Enteral

Examples of Enteral Access

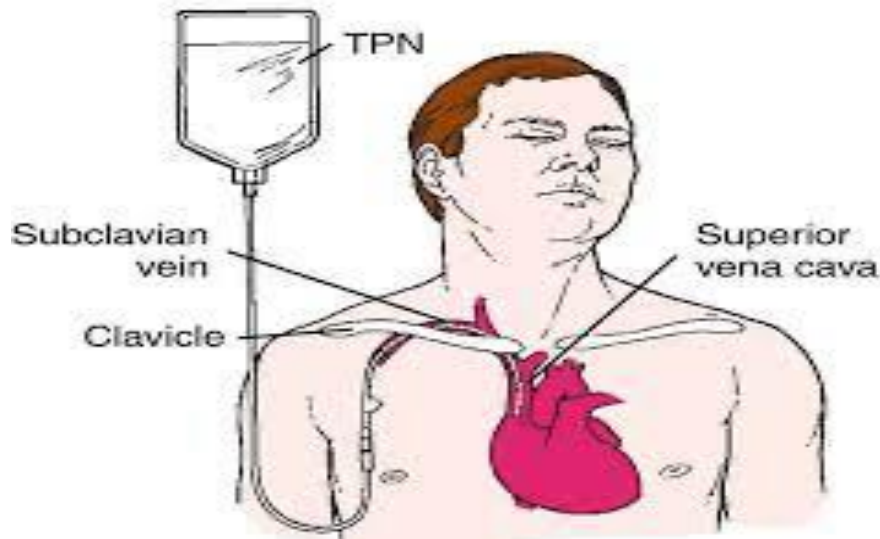


1. Check HCP prescription.
2. Assess bowel sounds.
3. Position semi-fowler to fowler.
4. Assess for placement
 - ✓ Aspirate gastric content
 - ✓ Check for PH = 3.5 – 4.
5. Warm feeding at room temperature.
6. Change the tube feeding container and tubing, every 24 hours
7. Check expiration date before hanging.
8. Use 50 – 60 ml syringe.
9. Gently flush
 - ✓ 15 – 30 ml before and after.
 - ✓ 5 ml between medication.

Complication of Enteral Nutrition

1. **Diarrhea** and abdominal cramp
 - ✓ **Slow feeding** administration and **warm**.
 - ✓ Change the formula.
2. **Aspiration**
 - ✓ Verify placement with X-ray.
 - ✓ Check aspiration content (every 4 hours)
3. **Dehydration**
 - ✓ Mild fever = 100.4 – 100.5 F
 - ✓ Flushing tube with water
4. **Hyperglycemia**
 - ✓ Slow infusion rate
 - ✓ Administer insulin

Total Parenteral Nutrition (TPN)



- | | |
|-----------------|---------------|
| - Carbohydrates | - Mineral |
| - Fact emulsion | - Electrolyte |
| - Proteins | - Water. |
| - Vitamins | |

Indications

- Client severely dysfunctional or nonfunctional GI tract.
- Bowel rest.
Ex: Pancreatitis, coma, shock septic, seizure.
- Administered through **Central Vein** (subclavian or external jugular).

- Avoid abrupt discontinuation to prevent rebound hypoglycemia.

Complication

1. **Infection – if fever or chills** (cvc infection)
 - ✓ Notify HCP
 - ✓ Remove Catheter
 - ✓ Send tip for culture
 - ✓ Antibiotic
2. **Rebound hypoglycemia**
 - ✓ Administered bag 10 % dextrose
 - ✓ Keep dextrose at bed side 10 % dextrose bag
3. **Air Embolism**
 - ✓ Shortness of breath
 - ✓ Tachypnea
 - ✓ Chest pain
 - ✓ Cyanosis
 - ✓ Hemoptysis

Intervention for Air embolism

1. Clamp catheter
2. Place client in Trendelenburg position over left side.
3. O2
4. Call Dr.
- 5.

Catheter embolism (trozo de plástico para dentro)

- ✱ **Peripheral line**
- ✱ **PICC line**
- ✱ **MID line**

- During withdrawal check for intactness.
- Place tourniquets proximal to catheter site.

4. Pneumothorax

- ✓ Skin crepitus
- ✓ Decreased or absent unilateral breath sounds

6. Hyperglycemia.

Surgical Drainage

Penrose	Jackson-pratt	Hemovac
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-Simple latex drain -Not sutured in place -Open Drainage -More risk for infection	-Abdominal surgery -Mastectomy -Breast surgery -Portable wound, self-suction device with reservoir -80-100 ml of volume, Empty when half full -Keep below waist *Squeezing bulb flat before connected	-Larger portable wound self-suction device with reservoir -Volume until 400-500 ml -Notify HCP if it suddenly increases or become bright red -Empty when full or every 8 h -Cleanse opening and plug with alcohol -Compress evacuator completely before connecting.
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